Static 3D GNSS Position Check

Job Details

Job Number	DEMO-0001
Job Description	Static GNSS antenna position check (synthetic demo data)
Client	Example Client Inc. (fictional)
Party Chief	J. Doe
Surveyor	A. N. Other
Site / Berth	Demo Harbor, Berth 0 (fictional)
Vessel	MV Sample Vessel (fictional)
GNSS Device	Demo GNSS Receiver #1
Date	2025-01-01
Vertical Datum	NAVD88 (demo)
Instrument Station	CP-01 (demo)
Instrument Easting	385000
Instrument Northing	3285000
Instrument Station Height	2.500
Instrument Height	1.550
Backsight Station	CP-02 (demo)
Backsight Easting	385120
Backsight Northing	3285160
Backsight Height	2.815

Coordinate Reference System & Grid Factors

Coordinate Reference System	WGS 84 / UTM zone 15N (EPSG:32615)
Instrument Station Latitude	29.68976052°
Instrument Station Longitude	-94.18863260°
Grid Bearing to Backsight	036° 52' 11.63"
True Azimuth to Backsight	036° 16' 51.96"
Grid Convergence	-000° 35' 19.67"

Point Scale Factor	0.99976317

Grid convergence sign: positive when grid north is east of true north (grid bearing = true azimuth - convergence).

Horizontal Check

UTC Time	Grid Bearing (DMS)	Obs Dist (m)	Grid Dist (m)	Calc Easting	Calc Northing	GNSS Easting	GNSS Northing	dE (m)	dN (m)	Linear Misclose (m)
01:00:00	143° 07' 43.93"	75.018	75.000	385045.001	3284940.001	385044.983	3284939.963	0.018	0.038	0.042
01:02:00	143° 07' 46.83"	75.018	75.000	385045.001	3284940.000	385045.004	3284939.997	-0.003	0.003	0.005
01:04:00	143° 07' 49.43"	75.019	75.001	385045.000	3284939.999	385044.994	3284939.990	0.006	0.009	0.011
01:06:00	143° 07' 47.03"	75.020	75.002	385045.002	3284939.999	385044.994	3284939.979	0.008	0.020	0.021
01:08:00	143° 07' 52.03"	75.016	74.998	385044.998	3284940.001	385045.012	3284940.010	-0.014	-0.009	0.017
01:10:00	143° 07' 48.03"	75.018	75.000	385045.000	3284940.000	385045.002	3284940.012	-0.002	-0.012	0.012
01:12:00	143° 07' 49.43"	75.018	75.000	385045.000	3284940.000	385044.986	3284940.009	0.014	-0.009	0.017
01:14:00	143° 07' 50.53"	75.013	74.995	385044.997	3284940.003	385044.964	3284939.999	0.033	0.004	0.033
01:16:00	143° 07' 51.03"	75.016	74.998	385044.998	3284940.001	385045.003	3284940.013	-0.005	-0.012	0.013
01:18:00	143° 07' 51.93"	75.018	75.000	385044.999	3284939.999	385044.982	3284939.986	0.017	0.013	0.022
01:20:00	143° 07' 49.03"	75.017	74.999	385044.999	3284940.000	385045.023	3284939.973	-0.024	0.027	0.036
01:22:00	143° 07' 50.43"	75.016	74.998	385044.998	3284940.001	385044.994	3284939.991	0.004	0.010	0.011

Observations entered	12
Successful	12
Rejected	0
Distance scale factor applied	0.99976317

Quantity	Count	Mean (m)	Std Dev (m)	RMS (m)	Max |value| (m)
Linear misclose	12	0.020	0.011	0.023	0.042
dE (calc - GNSS)	12	0.004	0.015	0.015	0.033
dN (calc - GNSS)	12	0.007	0.016	0.017	0.038

Vertical Check

UTC Time	Station Height (m)	Instrument Height (m)	Obs Vertical Dist (m)	Target Offset (m)	Calc Height (m)	GNSS Height (m)	dH (m)
01:00:00	2.500	1.550	4.563	0.120	8.733	8.754	-0.021
01:02:00	2.500	1.550	4.567	0.120	8.737	8.751	-0.014
01:04:00	2.500	1.550	4.560	0.120	8.730	8.808	-0.078
01:06:00	2.500	1.550	4.560	0.120	8.730	8.740	-0.010
01:08:00	2.500	1.550	4.567	0.120	8.737	8.729	0.008
01:10:00	2.500	1.550	4.566	0.120	8.736	8.732	0.004
01:12:00	2.500	1.550	4.562	0.120	8.732	8.747	-0.015
01:14:00	2.500	1.550	4.567	0.120	8.737	8.673	0.064
01:16:00	2.500	1.550	4.572	0.120	8.742	8.749	-0.007
01:18:00	2.500	1.550	4.562	0.120	8.732	8.736	-0.004
01:20:00	2.500	1.550	4.565	0.120	8.735	8.721	0.014
01:22:00	2.500	1.550	4.568	0.120	8.738	8.730	0.008

Quantity	Count	Mean (m)	Std Dev (m)	RMS (m)	Max |value| (m)
dH (calc - GNSS)	12	-0.004	0.032	0.031	0.078